Building a Scalable E-Commerce Data Pipeline

Turning Raw E-Commerce Data into Business Insights

BRAZILIAN E-COMMERCE DATASET OLIST 2016–2018

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The Problem We Solved

Data Fragmentation

A Brazilian e-commerce company (Olist) had raw transactional data across 9 separate files with no unified view.

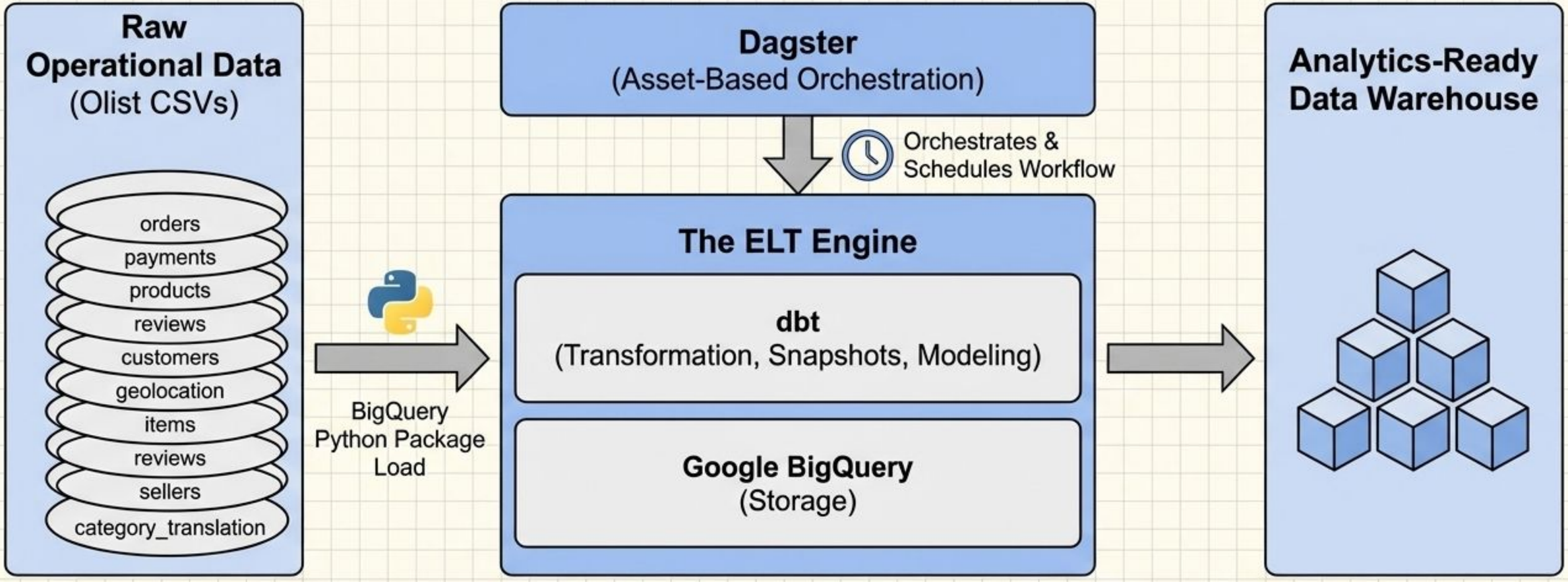
Lack of Insights

Business teams could not easily answer questions about sales trends, customer behaviour, or delivery performance.

The Solution

We built an automated, end-to-end data pipeline that ingests, cleans, validates, and analyses this data daily.

A Modern ELT Pipeline for Scalable E-Commerce Analytics



Automated Extract & Load
Raw operational files are loaded directly into BigQuery staging tables using the BigQuery Python package, bypassing fragile ETL bottlenecks.

Governed Transformation
dbt processes the staged data, employing snapshots to track changes and applying software engineering best practices like version-controlled SQL, structured data modeling, and rigorous quality testing.

Asset-Based Orchestration
Dagster manages the entire workflow execution order, scheduling, and dependencies, ensuring data freshness, idempotency, and real-time observability.

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Purpose-Built Machinery for Scale, Speed, and Governance

Google BigQuery
Cloud Data Warehouse

The Role
Serverless Compute & Storage (Project ID: academic-era-488315-j5).

The Business Choice
Decoupled storage and compute allows infinite scaling of queries without infrastructure management. Highly cost-effective for intermittent ELT workloads.

dbt Core
Transformation & Modeling

The Role
SQL-First Data Modeling and Automated Testing.

The Business Choice
Treats data like software. Replaces opaque drag-and-drop legacy tools with Git version control and verifiable data quality tests (dbt_expectations, dbt_utils) to prevent broken dashboards.

Dagster
Asset-Driven Orchestration

The Role
Pipeline Dependency and Execution Management.

The Business Choice
Unlike traditional task-based orchestrators, Dagster focuses explicitly on the data assets themselves. This provides unparalleled visibility into data lineage and faster troubleshooting.

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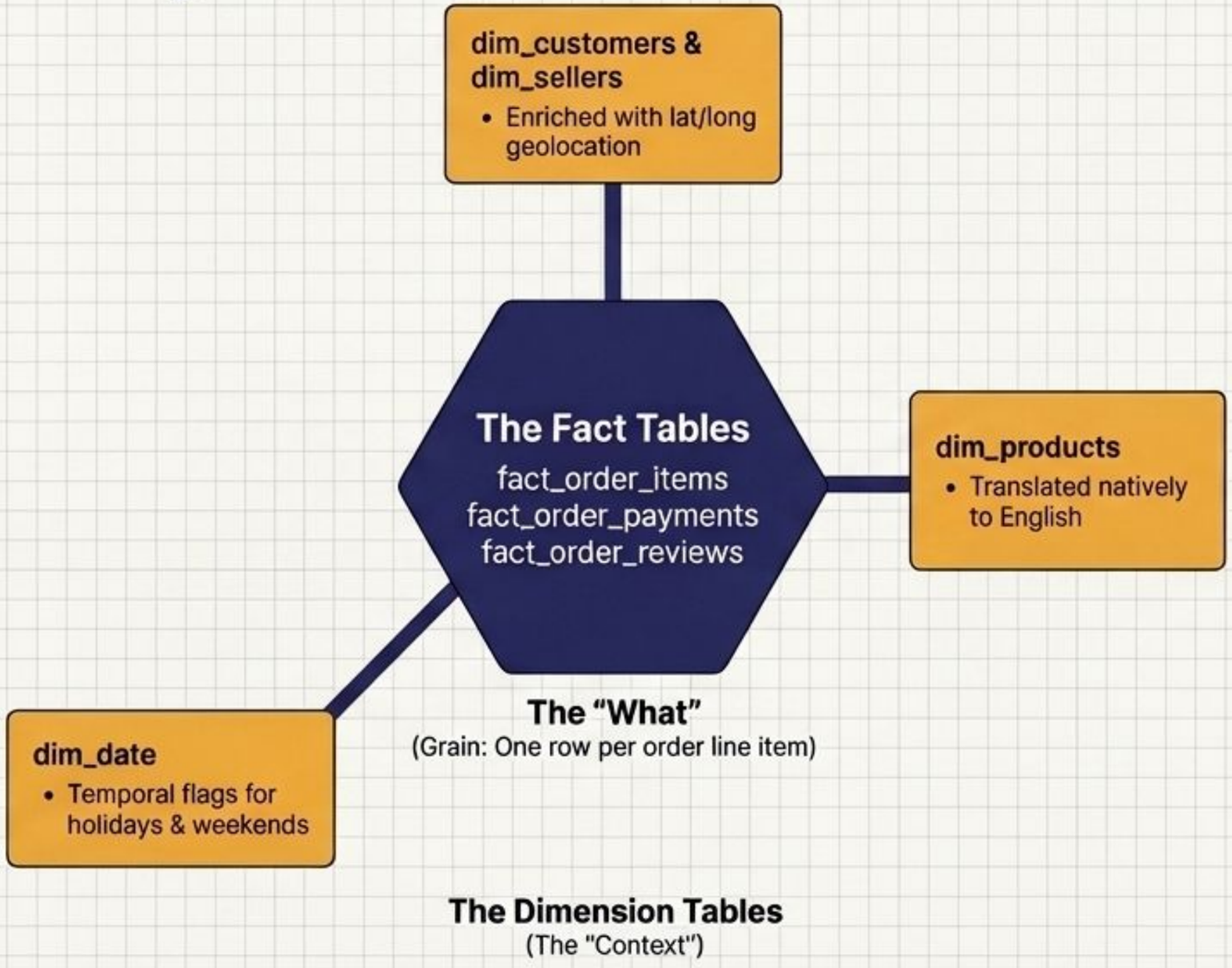
The Star Schema Optimizes for Analytical Performance



The Challenge

Raw operational datasets are highly normalized and require massive, slow database joins to answer simple business questions.

The Architectural Solution

A dimensional Star Schema flattens complex operational relationships into wide, easily scannable tables purpose-built for analytics.

The Business Velocity Impact

Lightning-Fast BI. Native compatibility with modern BI platforms means business users can instantly slice revenue by product category or seller location without writing complex 10-way SQL joins.

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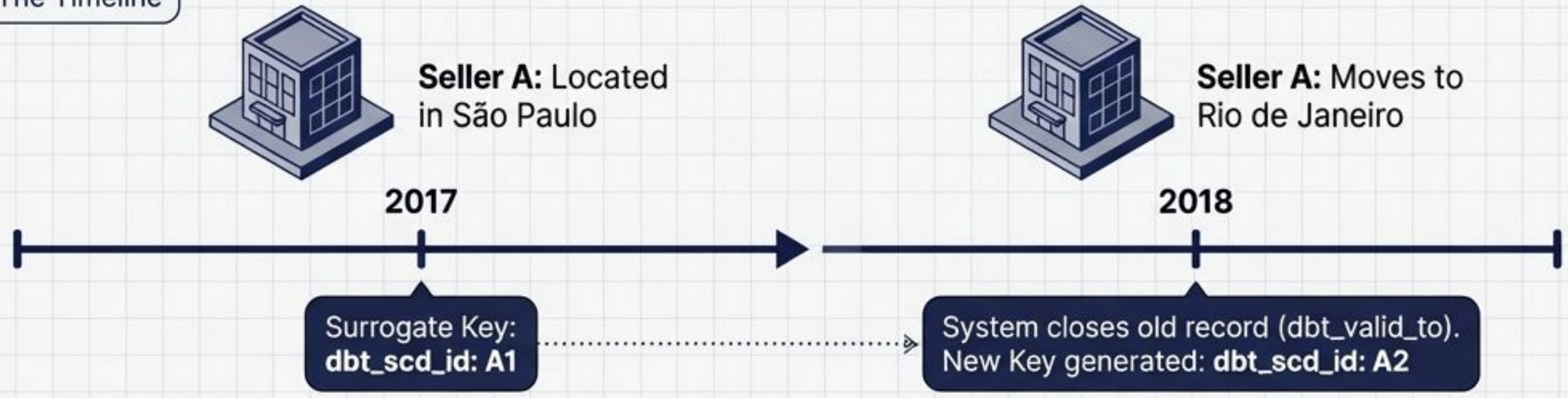
Architectural Accuracy Through Slowly Changing Dimensions

The Timeline



The Engine at Work

fact_order_items executes a temporal join on **order_purchase_timestamp** to lock in the exact profile that existed when the transaction occurred.

The Hidden Challenge:

Overwriting old addresses corrupts historical regional sales data, ruining year-over-year geographical analyses.

The Pipeline Solution:

dbt snapshots track state changes. Hard deletes are safely invalidated.

Surrogate keys separate history from the present.

The Business Result:

Absolute historical accuracy. An order placed in 2017 reflects the reality of 2017.

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dagster

Overview
Runs
Catalog
Jobs
Automation
Lineage
Deployment

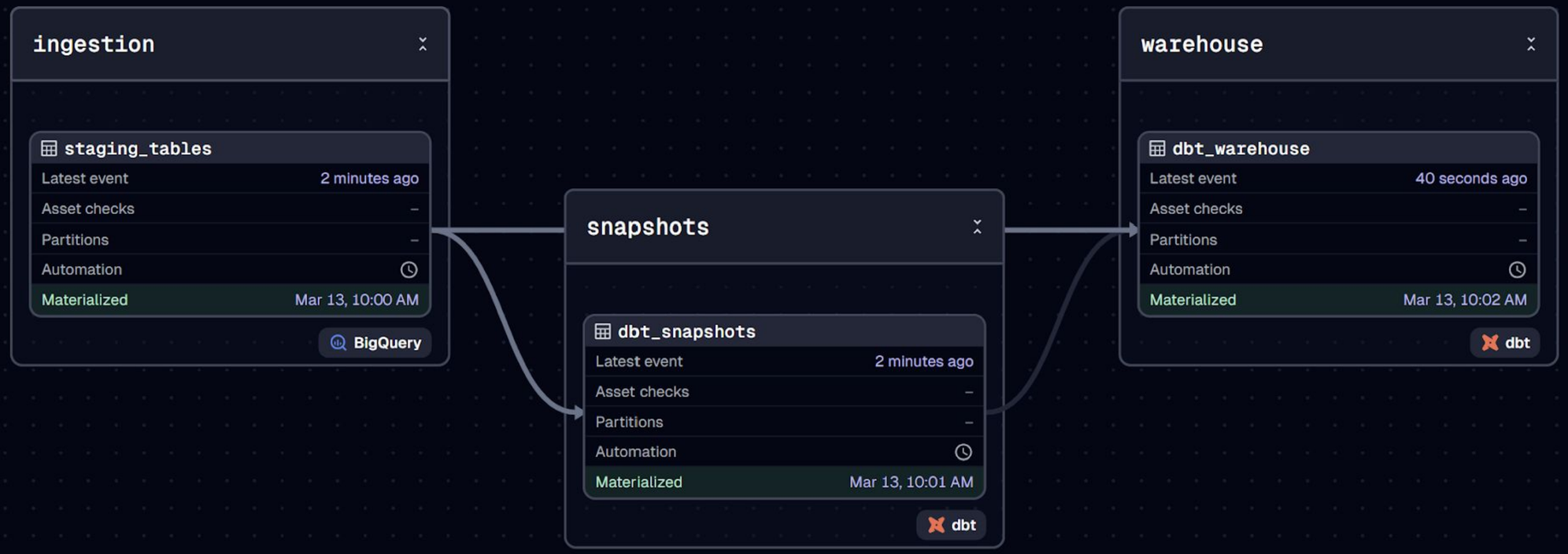
Search
Hide navigation
Settings
Support

Global Asset Lineage

Reload definit

Search and filter assets

Materialize al



The Olist Ecommerce — Everything in One View

A dashboard grid of key metrics for Olist.

99,441
Total Orders

BRL 15,843,553
Total Revenue

96,096
Unique Customers

November 2017
Peak Revenue Month
Highest revenue month in the Dataset

Health & Beauty
Top Revenue Category

Bed, Bath & Table
Highest Volume Category

Credit Card (73.9%)
Top Payment Method

São Paulo (SP)
Top Revenue State

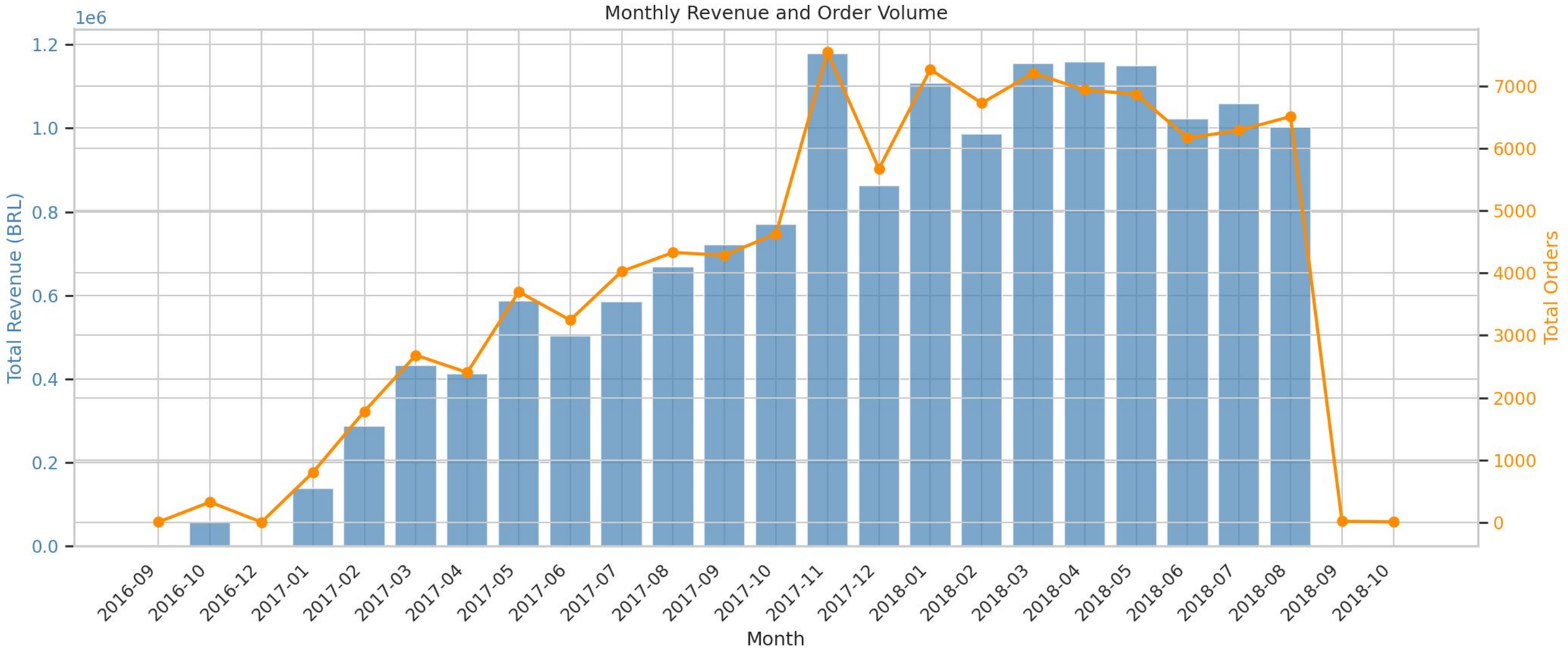
BRL 201
Highest Avg Order Value
Watches Gifts

4.09 / 5.0
Average Review Score

96.9%
One-Time Buyer Rate
Customers who never return

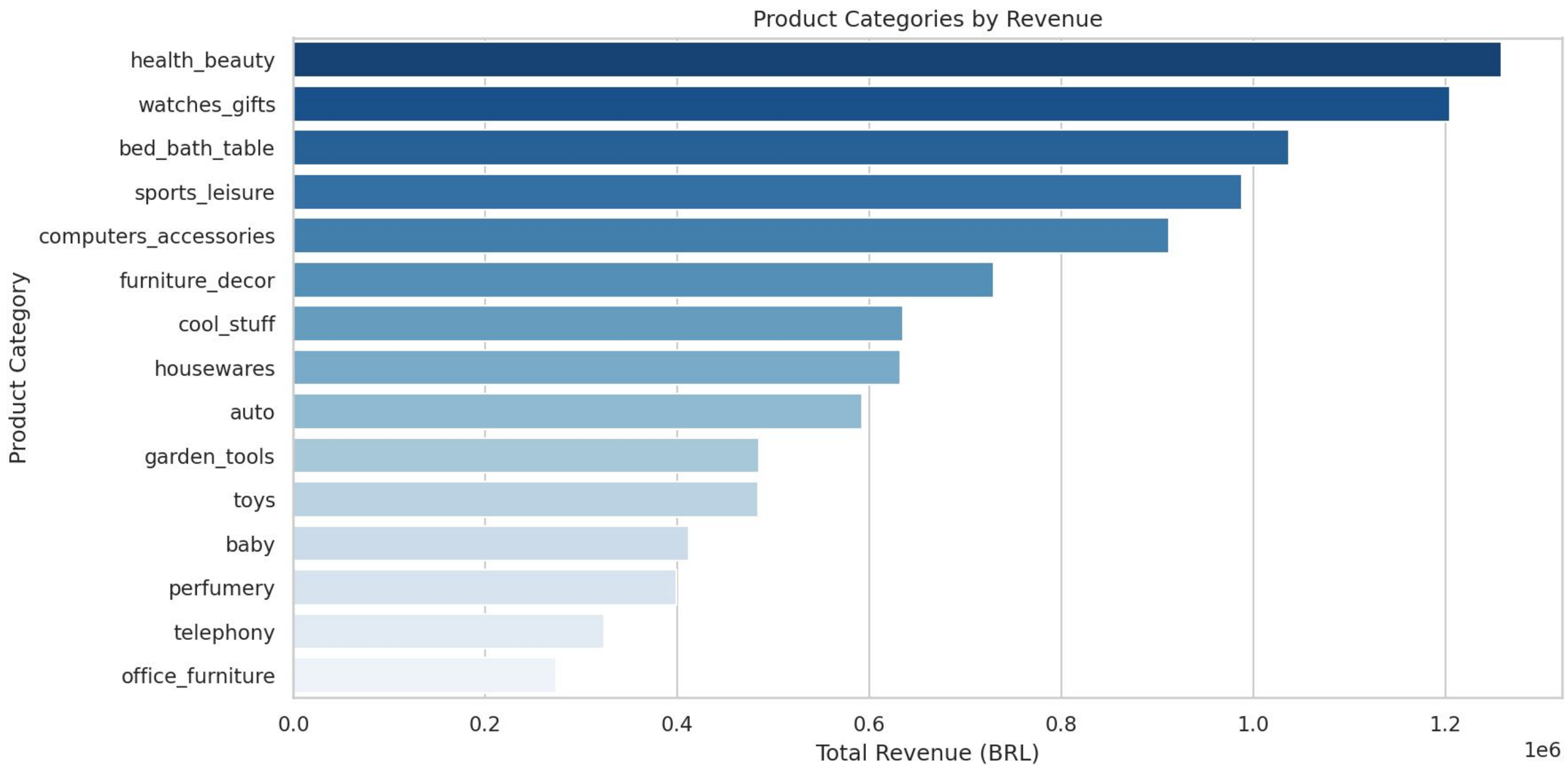
12.5 days
Avg Delivery Time — National
SP average is 8.7 days

Revenue Grew 10× in 12 Months - Consistent Growth Across 25 Months of Data



- Revenue and order volume grew steadily from 2016 to late 2017, peaking in November 2017.
- There is a sharp drop in late 2018, which reflects incomplete data for that period rather than an actual decline.

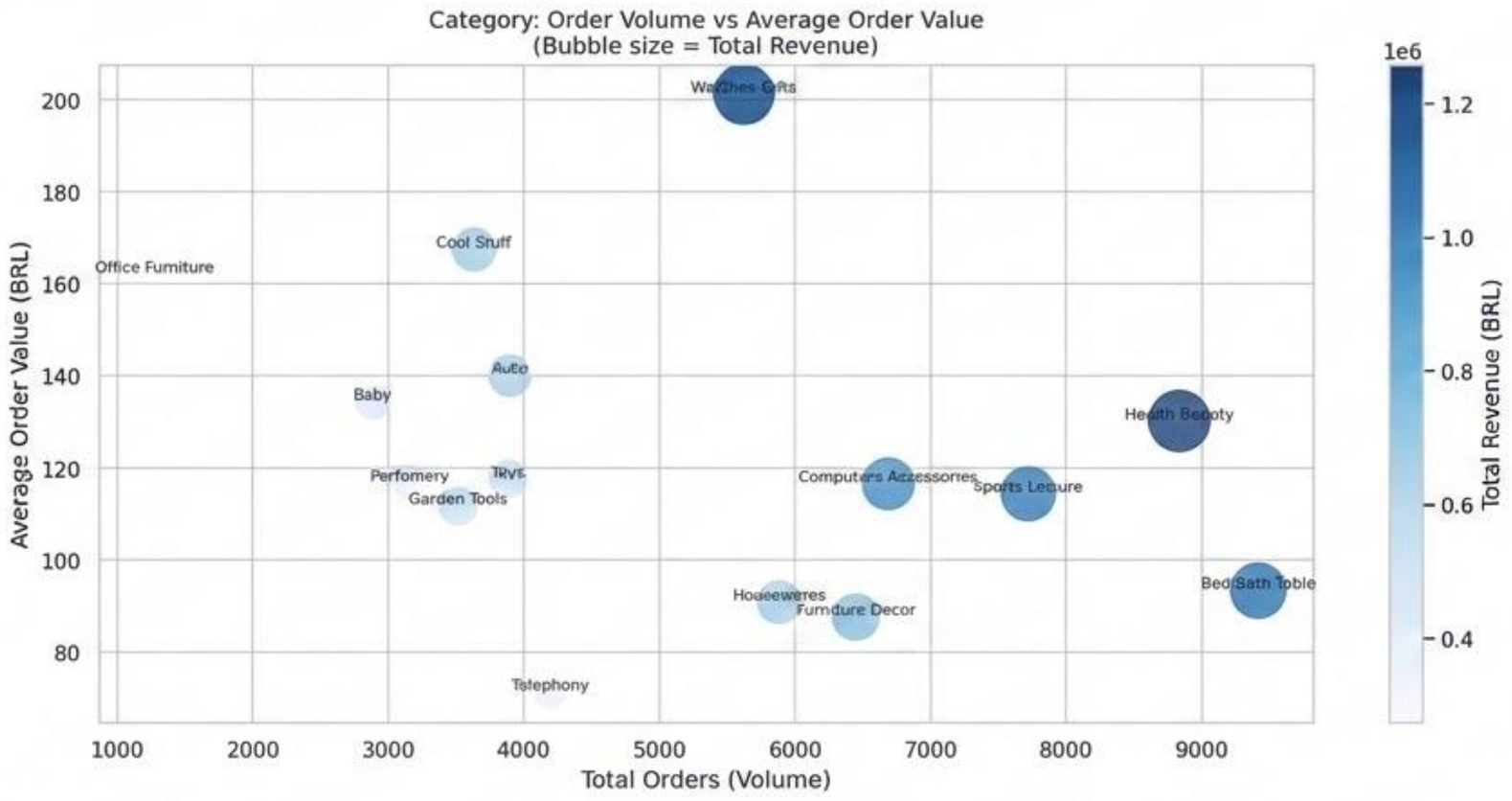
Top-selling Products by revenue



- Health & Beauty leads in both revenue and consistency

Category Revenue VS Untapped Potential

Beyond identifying the top revenue category, we examine which categories have high average order values but low order volumes. These represent untapped commercial opportunities.



Category	Total Orders	AOV	Total Revenue
health_beauty	8836	BR 130	BR 1,258,861
watches_gifts	5624	BR 201	BR 1,205,005
Bed_bath	9417	BR 93	BR 1,036,988
Cool_stuff	3632	BR 167	BR 635,290
office_furniture	1273	BR 162	BR 273,960

- **Bed & Bath** – highest order volume, but relatively low AOV, suggests price-sensitive buyers.
- **Watches & Gifts**, Office Furniture, and Cool Stuff show premium buying behaviour at lower volumes, signalling untapped market potential.

São Paulo Leads in Volume - But Northern Customers Spend More Per Order

Volume leader

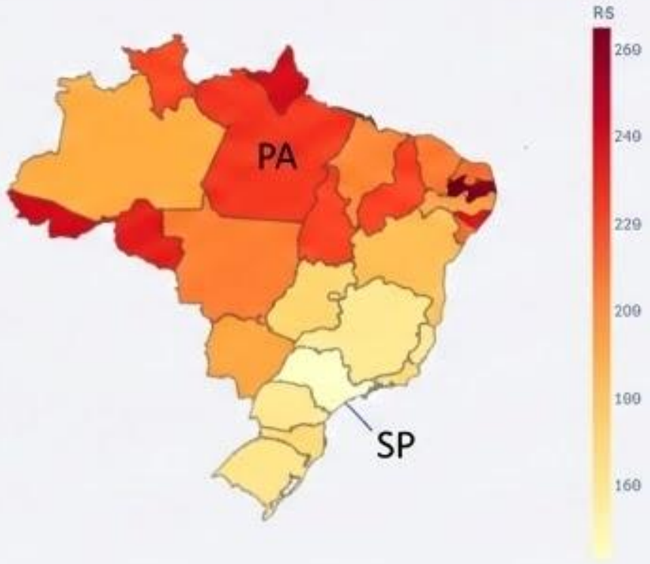
São Paulo (SP) dominates total sales

Hidden opportunity

Northern states like Pará (PA) have higher per-customer spending than São Paulo. Signal customers here are more willing to spend

Sales by State in Brazil - Total Sales (R$)

Sales per Customer (R$)



The most valuable customers are not always where the most orders come from.

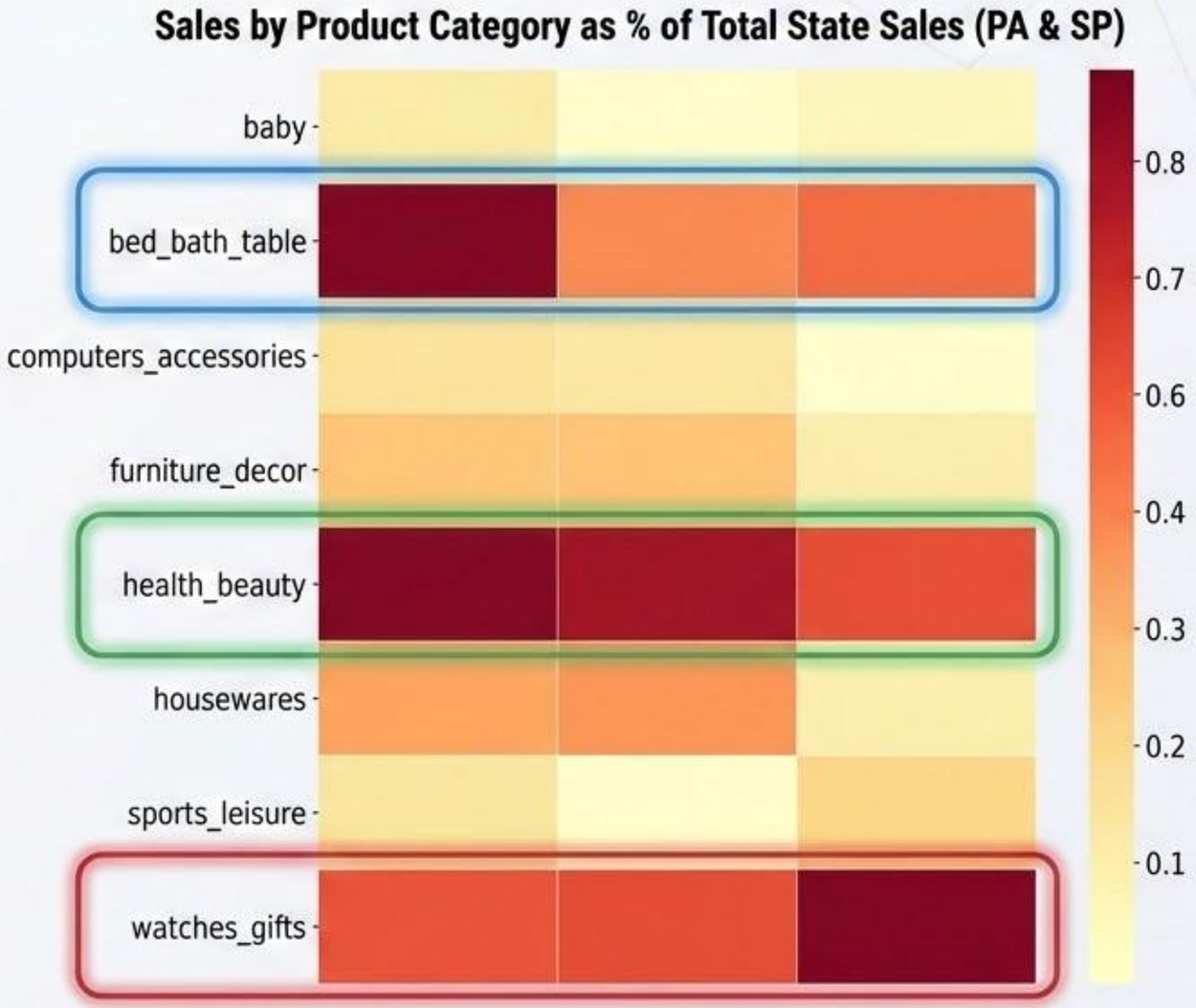
SP Has the Volume, PA Has the Value

Health & Beauty is the strongest category in PA, commanding a larger share of state sales than any other category, and proportionally stronger than in SP. Demand for this category runs deeper.

Watches & Gifts is disproportionately stronger in PA than SP. PA customers allocate a noticeably higher share of spending to this premium, high-value category compared to SP customers.

Bed, Bath & Table dominates SP but is significantly underrepresented in PA — despite PA customers spending more per order overall. This gap suggests unmet demand.

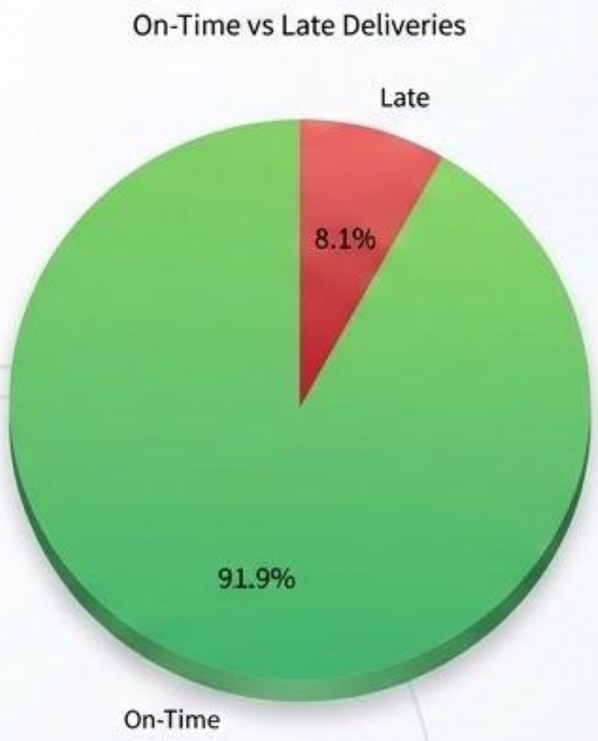
This is not a struggling market. It is an undiscovered one.

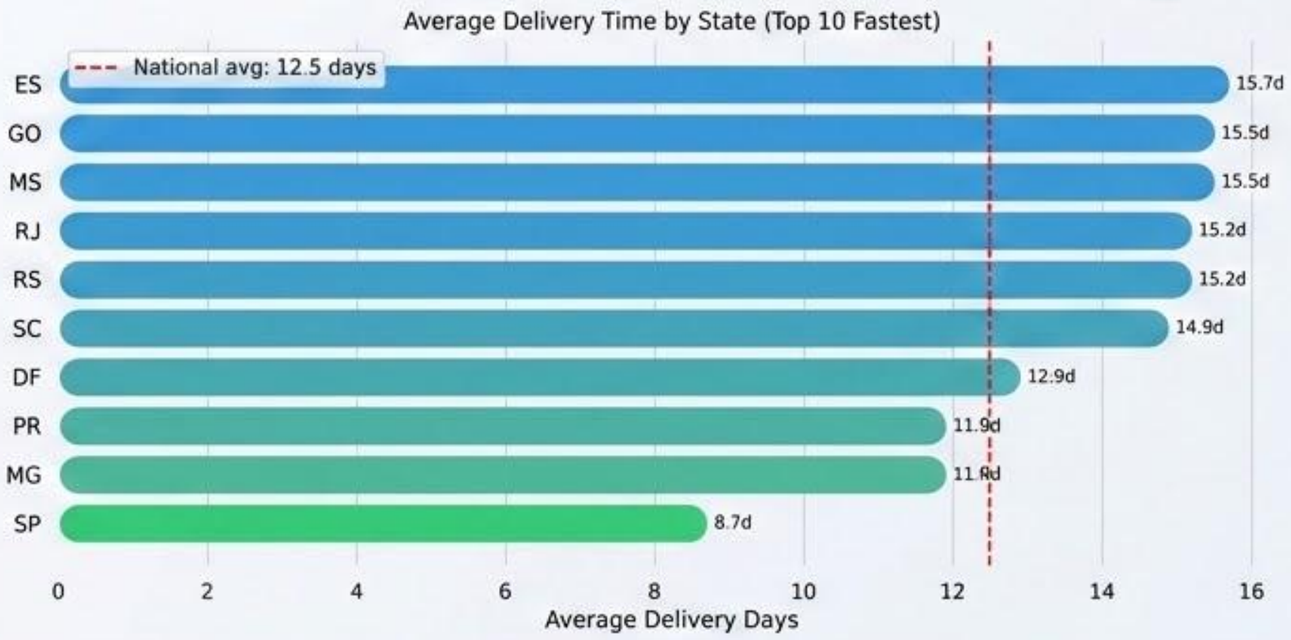


Delivery Performance

91.9%
On-time delivery rate

12.5 days
National avg delivery time
(SP avg is 8.7 days)





- States outside the southeast corridor average 14–16 days. These are the states generating the most 1-star reviews.

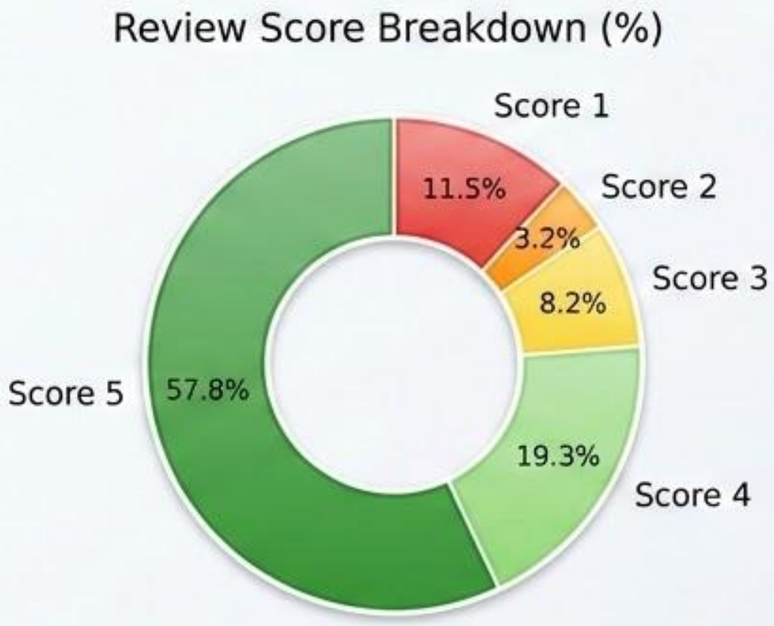
Customer Satisfaction

11.5% Give 1-star reviews

1 in 9 customers

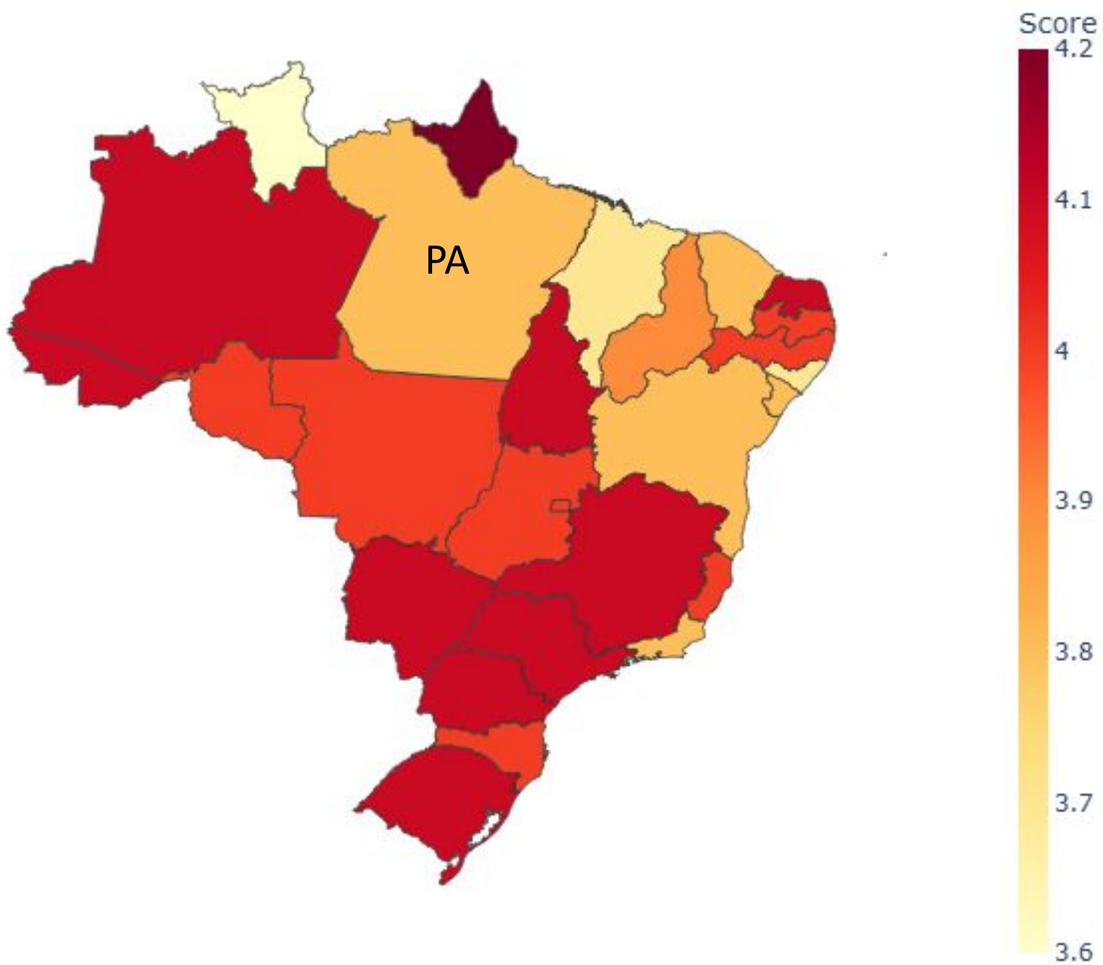
57.8% Give 5-star reviews

When it works, it works



When delivery is fast, customers give 5 stars. When it is slow, they give 1 star and never return. This directly drives the retention crisis

Satisfaction Is Lower in PA, but the Opportunity Is There



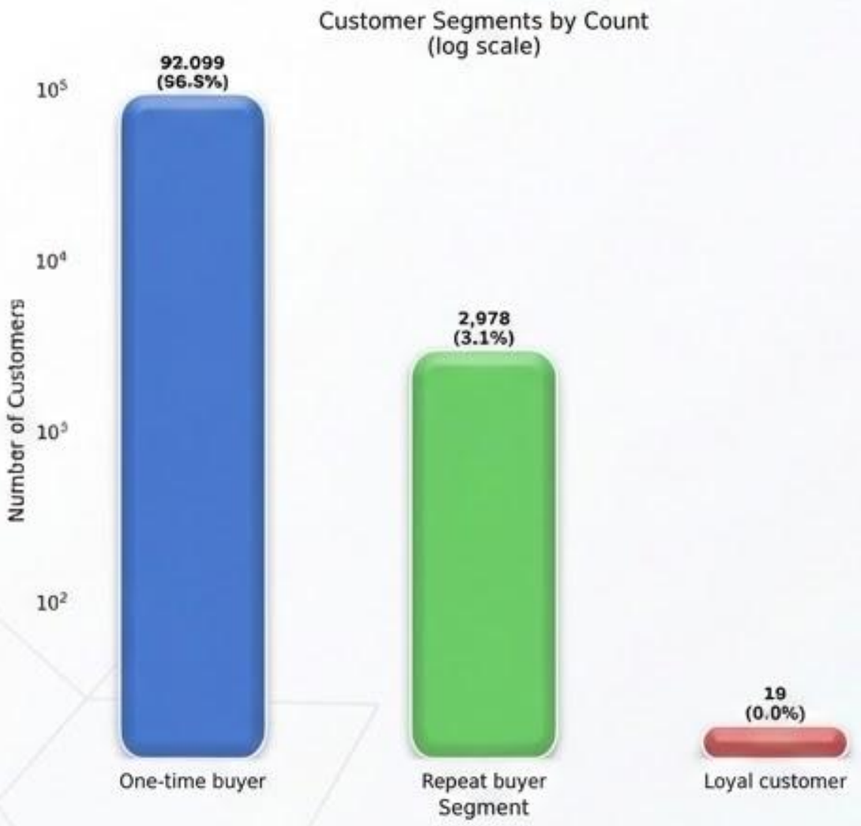
SP scores 4.18 — among the highest in Brazil, and delivered in just 8.7 days. Fast delivery directly drives high satisfaction in the southeast.

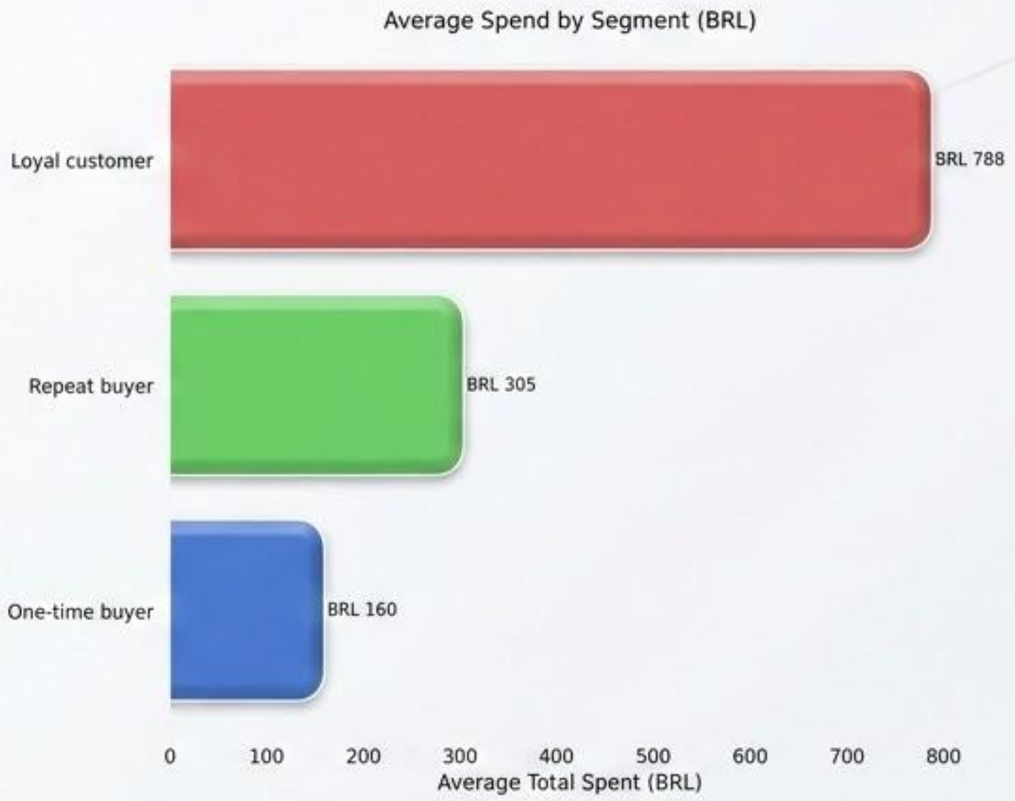
PA scores 3.85 — below the national average, and takes 23.6 days to deliver. The lower score is due to the long delivery time.

Neighbouring AM scores 4.18 despite 26-day delivery times — matching SP's score. This shows northern customers are not inherently dissatisfied. Fix delivery in PA, and satisfaction will follow

- **PA's lower score is a delivery problem. Solve logistics, and the north becomes as satisfied as the south.**

96.9% of Customers Buy Once and Never Return





Key Insights

Retention Crisis: 96.9% of customers purchase only once.

High Value of Loyalty: The few loyal customers (19) spend avg. BRL 788 vs. BRL 160 for one-time buyers.

Opportunity: A loyalty program or targeted campaign for one-time buyers could significantly increase revenue.

Three Actions to Turn Growth Into Lasting Revenue

Build Customer Loyalty

Problem: 96.9% of customers never return.

Solution: Launch a loyalty programme — rewards, personalised offers, and repeat-purchase incentives for existing customers.

Impact: Loyal customers spend BRL 788 vs BRL 160. Converting 5% of one-time buyers = significant revenue uplift at zero acquisition cost.

Fix the Delivery Experience

Problem: 12.5-day waits are generating 1-star reviews and lost customers.

Solution: Set maximum delivery time standards for all sellers. Prioritise logistics improvement on slow routes outside São Paulo.

Impact: 11.5% of customers give 1 star. Eliminating the 30–60 day delivery tail would directly improve retention.

Expand into the North

Problem: High-value northern customers are underserved.

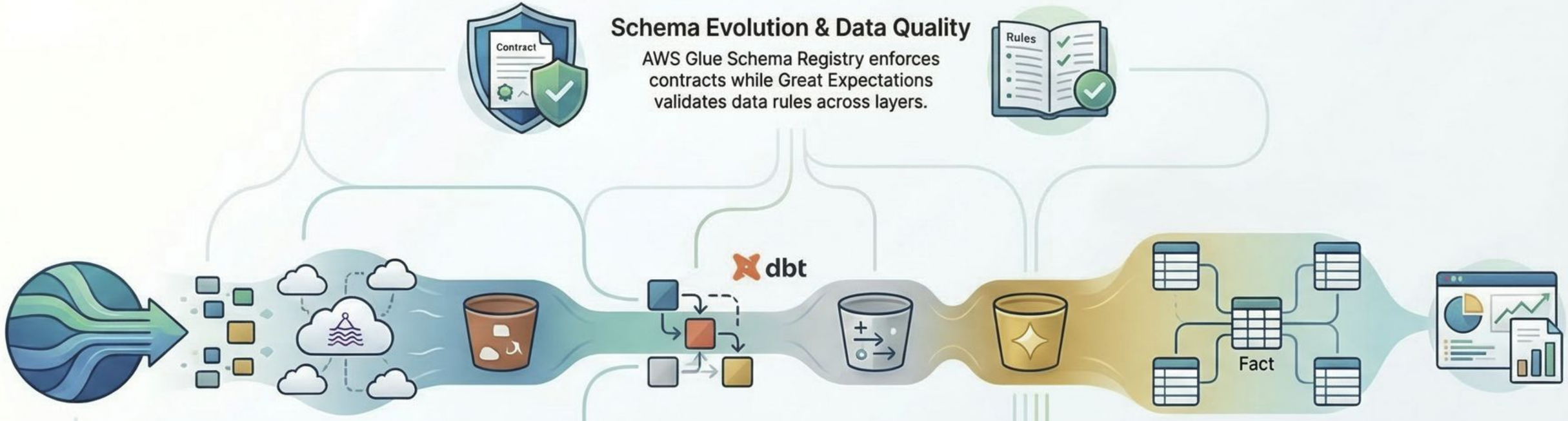
Solution: Recruit sellers in Pará and neighbouring northern states. Invest in northern logistics partnerships.

Impact: PA customers 57% more per order as compared to SP. Fix delivery and satisfaction will follow, making this the lowest-risk expansion market available

Every number behind these recommendations comes from real transaction data — 99,441 orders, 96,096 customers, 25 months

APPENDIX

Keystone Nexus: Engineering a Real-Time Medallion Data Pipeline



Schema Evolution & Data Quality

AWS Glue Schema Registry enforces contracts while Great Expectations validates data rules across layers.

Real-Time Streaming Ingestion

Data flows from producers through AWS MSK and MSK Connect into the Bronze (Raw) S3 layer.

Multi-Stage Transformation

dbt cleanses data into the Silver layer before curating it into the Gold layer.

Optimized Gold Star Schema

Final curated tables are structured into Fact and Dimension tables optimized for BI consumption.

Compute Elasticity & Performance

AWS Auto Scaling responds to Kafka lag, while Athena partitioning reduces query costs.

Connection Resiliency

Amazon RDS Proxy manages connection multiplexing to ensure database stability and failover resilience.

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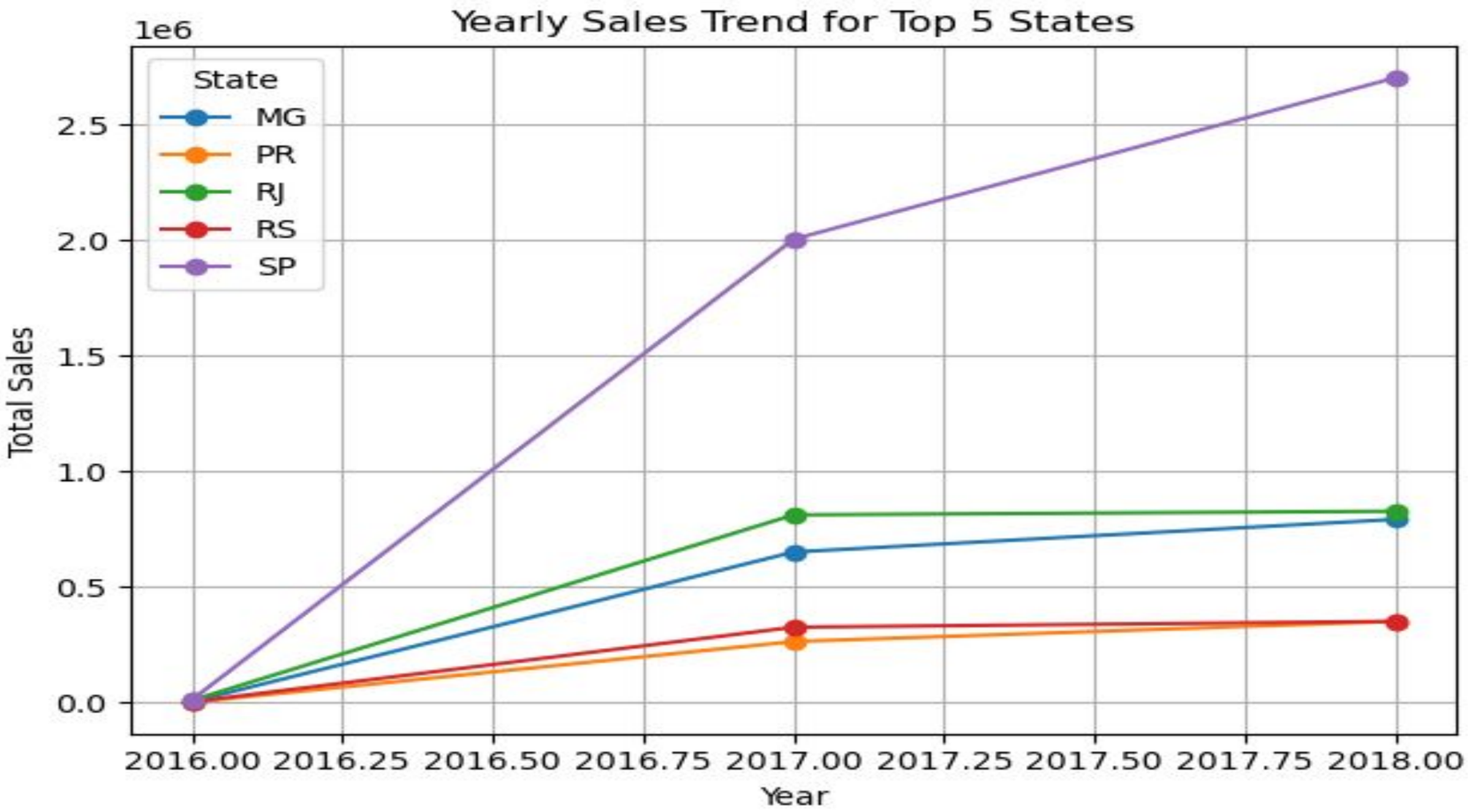
Top - Yearly sales by States



State SP leads in yearly total sales, Graph shows every quarterly sales trends in top 5 states